TASK 3 — Research Results, Data Analysis, and Scientific Visualization

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Research Methods · Week 7–9

Research context: Trust of students and patients in AI-generated medical recommendations. This submission benchmarks the **survey data-processing pipeline**, not an AI medical recommendation model.

1. Methods and provenance

The source sample is a 10-row synthetic survey. The Week 7 log preserves 120 technical measurements; Task 3 adds four fixed workloads (1, 10, 25 and 50 calculations per measurement), with 120 timed observations at each load. Ten untimed warm-up calls per workload were discarded prior to JSONL logging; console output and disk writes were kept outside the timed section. These are locally measured engineering timings, not clinical findings.

2. Statistical analysis

We compute median total latency, p95/p99 tail latency, median latency per calculation, bootstrap percentile 95% confidence intervals (2,000 resamples; fixed random seed 42), and error rate including unsuccessful measurements in the denominator.

Workload	Median total (µs)	p95 (µs)	p99 (µs)	Per calculation (µs), 95% CI	Error
1	13.8	16.8	42.1	13.80 [13.72, 13.90]	0.0%
10	131.7	279.4	690.2	13.17 [13.12, 13.28]	0.0%
25	343.7	461.9	610.6	13.75 [13.39, 13.91]	0.0%
50	700.4	987.1	1153.4	14.01 [13.64, 14.63]	0.0%

3. Discussion

The baseline is one calculation; the proposed batch workload is 50 repeated calculations using the **same algorithm and synthetic dataset**. This is a workload comparison, not an AI method comparison. Total elapsed time grows with load. Per-calculation timings and percentile tails reveal the cost distribution; bootstrap intervals describe uncertainty conditional on this experimental environment. No participant-level trust hypotheses were tested.

4. Scientific figures

Figure 1. Latency per calculation by fixed workload with bootstrap 95% confidence bars.

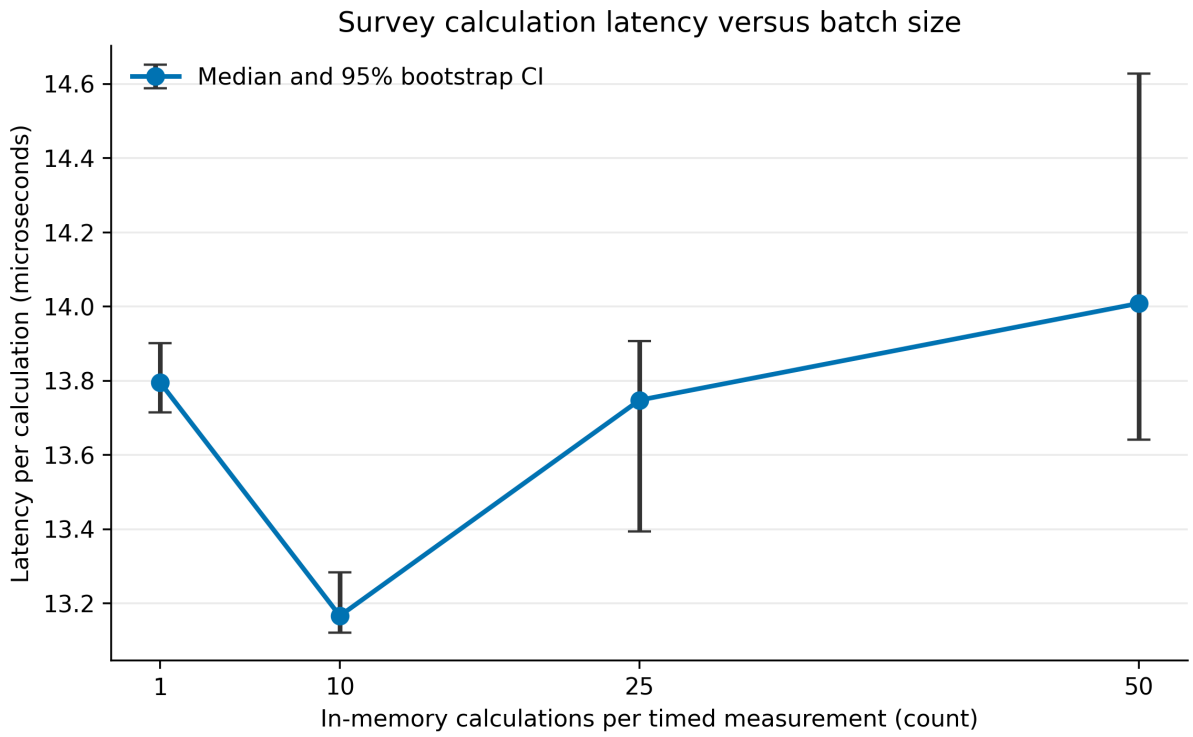
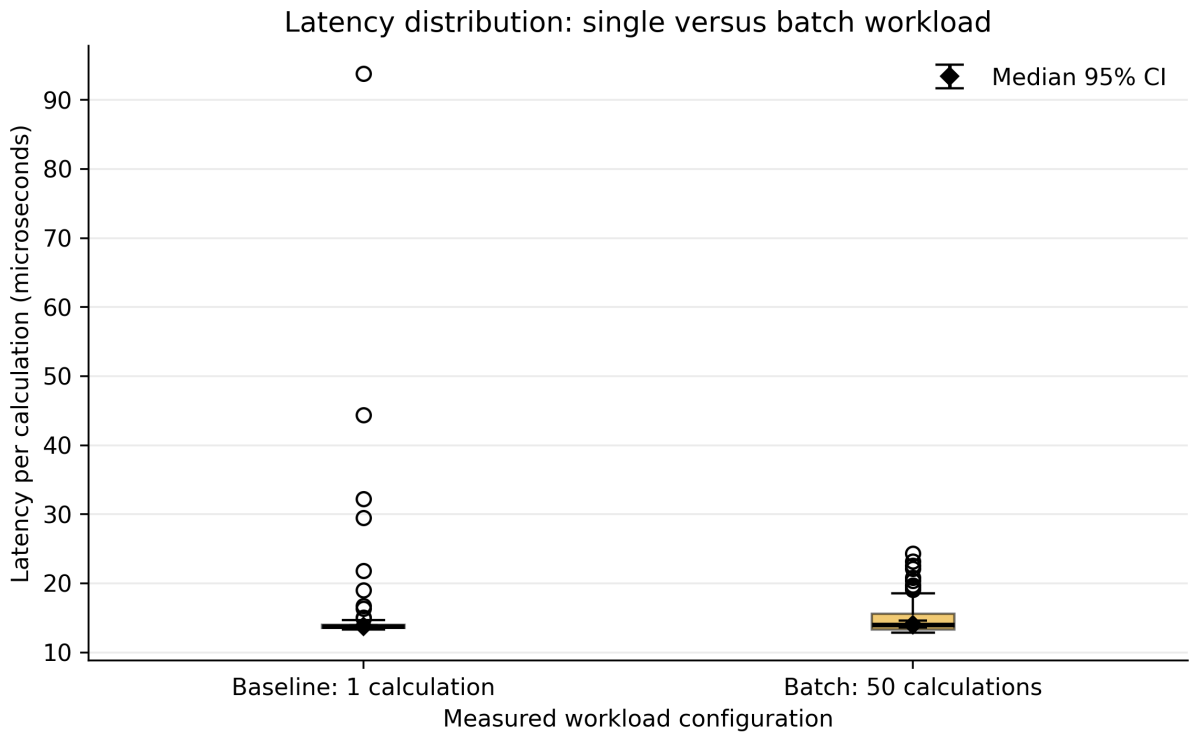


Figure 2. Run-to-run latency distribution and uncertainty for baseline versus 50-calculation workload.



5. Domain model and reproducibility

The C4-style Mermaid component model in **docs/domain_model.md** identifies the CSV input, in-memory computation, collector measurement point, JSONL log, statistical aggregation, and scientific plots. The Jupyter notebook is executed and preserves all outputs. Figure files are committed as 320 DPI PNG and vector PDF; the original Task 2 files are preserved.

Limitations: only synthetic responses are used; benchmark timings are local-machine measurements, warm-up timings were excluded rather than saved as observations, and no medical or population-level conclusions can be drawn.

Repository: https://github.com/marzhanolzhabek/ai-medical-trust-research